

Bell Peppers (Field)

CULTIVARS

- Carmen
- Lipstick
- Ace
- Sprinter
- Escamillo

INTENSIVE SPACING

- 2 rows per bed spaced every 12" (30 cm) on the row.

FERTILIZATION

- Field bell peppers are heavy feeders that stay in the garden for 100 + day . We front-load the total amount of fertility requirement at the initial bed prep stage and do not top dress them afterwards.
- Apply lots of compost, +/- 1.5 inch (7 cm) all across the bed along with a dose of nitrogen-rich organic fertilizers like pelleted chicken manure (5-3-2) and alfalfa meal (2-0-2). Also, add potassium sulfate (0-0-50) and organic Sul-Po-Mag (0-0-22)

IRRIGATION

Irrigation Type

- Drip tape (4 per bed) : Under plastic mulch.
- No sprinkler to keep foliage dry and thus prevent diseases.

Irrigation Management

- Root depth: Medium (24"-40")
- General: Constant and even moisture from flower through fruit. Peppers like to dry down before being watered again.

- In hot conditions, imbalances of moisture may lead to blossom end rot. To mitigate, frequent deep water is critical.

STARTING SEEDS

- Start transplants 7-8 weeks before transplantation date. Sow seeds in 128 cell flats in seed-starting soil mix. 1 seed per cell.
- Place trays on heating mats and maintain floor temperature at 80 °F (27 °C) until emergence.
- After emergence, maintain the ambient temperature in the nursery around 75°F (24 °C) during the day and 65°F (18 °C) at night. The soil-mix must be kept moist at all times.
- Pot-up in 4" (10cm) pots 3 weeks after sowing.
- Plants are ready to transplant when the seedlings are about 7-8 weeks old they should be 6-8" (15-20 cm) tall. Ideally, they will have some buds but no open flowers.
- Harden off plants by exposing them to outside temperatures and reducing water for 4-5 days before transplanting.

IMPLANTATION

Bell peppers are planted once a year and are heat-loving crops that like to be grown under cover (tunnels, caterpillars,). Pruning them one time will ensure a regular supply of fruits all the way to the end of the season.

EQUIPMENT

- rotary harrow
- broadfork
- 30" bed prep rake
- Landscape fabric, staples, drip tape and fittings.
- 4' (1.2M) steel bars (cement rods); 1 stake every 4 plants.

TRANSPLANT ITINERARY

- Before transplanting, make sure minimum soil temperature is above 60°F (16°C), 20°C is optimal.

- Pass through with the broadfork.
- Remove large debris from the bed if needed (use flail mower, rake or manually remove the large chunks).
- Apply compost and amendments in the middle of the bed and pass with the rotary harrow at a depth of 2.5" (6 cm). Since the bed is covered with landscape fabric, it needs not to be clean up with another pass of the rake.
- Install 4 lines of drip tape per bed and start the system. Fix leaks at that point, if any.
- Lay down landscape fabric and secured with staples each second step (5' or 1.5 m). It needs to be tight.
- Transplant bell peppers making sure to extract the seedlings from their pots delicately. When putting them in the ground, the soil level on seedling should end up being quite level with the soil.
- Using a sledgehammer, drive stakes 12' (30 cm) deep into the ground every 5' (1.8M).
- Water the bed thoroughly the first time, keeping the soil moist for the first week after the transplant.
- Once the plants have reached +/- 16" (40 cm) high, install the first line of bailing twine, crossing it from the inside-out of each stake. Basket-weaves the plants 2 or 3 more times as they grow every 8" (20cm).

WEEDING

- Weed through the holes of the landscape fabric 2 or 3 times in the season.

MAINTENANCE

- To maximize productivity, prune off the first pepper (flower) located at the first node (the V, where the plant branches out in two). Also, remove all the leaf from the main stem (before the first V). This will help the plant further develop its root system so allowing it to bare a more significant fruit load.
- When the plants reach its third node third (third V), prune off one branch of the two, the interior one. This pruning will help better distribute the fruit load of the plant.

PLANT PROTECTION

- Young bell pepper plants are susceptible to aphids. To cope, remove leaves that are heavily affected and introduce ladybugs or *Aphidius Colemanii*. If the situation is leaning towards infestation, spray your plants with insecticidal soap prior to introducing your beneficials.
- If plants are flowering but fail to set fruit, the culprit (at least in northern regions) is likely to be the tarnished plant bug, which particularly appreciates the sap of eggplant and pepper flowers. In the case of an infestation, use the recommended organic-farming dosage of *Spinosad*.
- To counter blossom-end rot (a physiological disorder usually related to calcium deficiency in the water), water abundantly during the fruit-formation period and especially during the fruit-maturation period. Aim for up to 1" (2.5 cm) water/day (including rain). In the case of rapid degradation, or to prevent the problem, spray your plants twice a week with calcium chloride. If all fails, fertigate the calcium directly in your drip system.

HARVEST

- Harvesting is done twice a week using a pruner and tree planting bags.
- Peppers are ready to harvest when they are firm, large and totally ripe (ie. fully colored).
- If not being sold right away, harvest peppers when they are 80% or more final ripe color. They will reach full color in storage.
- Only peppers that are free of blemishes are harvested. The rest are picked and discarded in the pathways.

CONDITIONING

- Bell peppers are not washed but must be wiped clean if dirty.

STORING

- Bell peppers are stored in cold room kept at 47 °F (8 °C) and should not be stored for more than 4 days before being sold.